JC Parks and Rec Technical Report

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Johnson City Parks and Recreation Economic Impact Analysis

Abstract

This study aims to perform an Economic Analysis on behalf of the Johnson City Parks and Recreation Department. Our partner, the Parks and Recreation Department of Johnson City, wants to know what the effect on tourism a planned park expansion will have on the region. The park expansion will consist of 4 new baseball/softball fields, and 6 soccer fields. This will bring the parks total amenities to 9 baseball/softball fields and 8 soccer fields. It is also important to note that the expansion will consist of entirely artificial play surfaces. Being that the park expansion will be an additional input for the regions tourism numbers overall, the parks department naturally wants to know what the expected economic impact will be on the region due to this increase in tourism. To perform this analysis we will be relying on information from a variety of sources in order to construct a table of known expenditures. From there, we will make use of the RIMS II multiplier tables which are generated using an I/O model of the region.This will allow us to create region specific impact estimates that we can share with the parks department, and they in turn can use to share with interested parties. This will also allow the park department to have estimates as to how this project will impact their bottom line.First and foremost we will be discussing the data as we received it from the parks and rec department. Second, we will cover the secondary studies used as reference for our own. Third we will cover preliminary calculations made using methods outlined in these secondary studies. Fourth, we will be talking about the Input/Output models and the RIMS II tables which will be utilized later in the study. And lastly, fifth, we will be talking about where the project will be going from here.

Part 1 The Data

Initially for this study we were presented with minimal information from the parks department itself. We received data for the 2022 and 2023 baseball seasons and the maintenance charged for the use of the baseball fields. The data was in the form of a word document and only contained the number of teams, the date of the tournament, and the amount charged to the director of the tournament for use of the fields. This was less than what we had hoped to get our hands on, but the none-the-less it was a start. To begin with, we took the data received and put it into the form of a table so that it was more readable in a raw format as well as making it compatible with software packages like R, Python, or Eviews. Additionally, we referred to some secondary studies in order to pad out the data further with such information as initial party spending estimates or non-local percenatage of attendees. We will cover this secondary information in Part 2, but for now, our focus will remain on the initial data.

```
## New names:
## * `` -> `...11`
## * `` -> `...12`
```

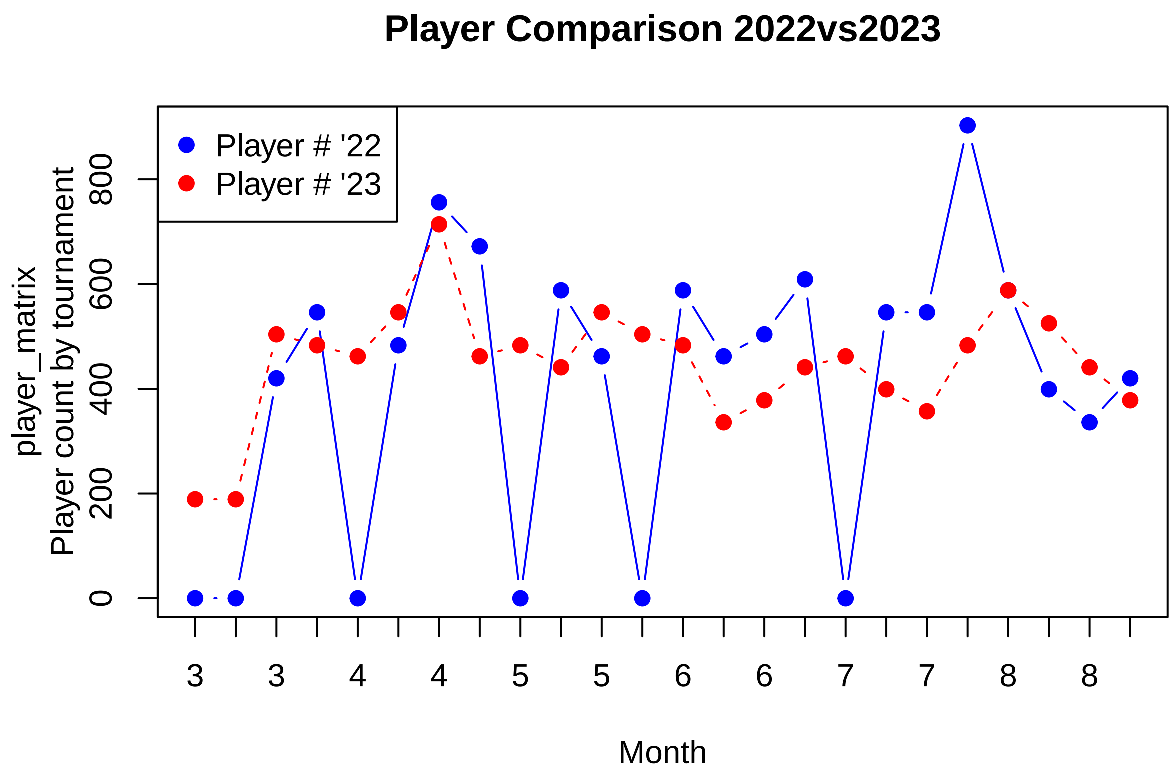
```
## # A tibble: 6 x 13
##    days teams   fee labor material combined month players spending nonlocal
##   <dbl> <dbl> <dbl> <dbl>    <dbl>    <dbl> <dbl>   <dbl>    <dbl>    <dbl>
## 1     2    29   820  3106      369     3475     3     609   250908  223308.
## 2     0     0     0     0        0        0     3       0        0        0
## 3     0     0     0     0        0        0     3       0        0        0
## 4     1    20   600  1326      219     1545     3     420    86520   77003.
## 5     2    26   900  3106      369     3475     4     546   224952  200207.
## 6     0     0     0     0        0        0     4       0        0        0
## # i 3 more variables: ...11 <lgl>, ...12 <lgl>,
## #   `Note: Cells 2-36 are 2022 data, 37-64 2023` <lgl>
```

A brief breakdown of the data set created so far:
* days: The duration of the tournament in days
* teams: The number of teams participating in a tournament
* fee: The fee charged to the tournament director for use of the park
* labor: The labor cost associated with running the tournament
* material: Material cost associated with running the tournament
* combined: Combined Labor and Material cost
* month: the month in which the tournament took place
* players: The number of players participating in the tournament (extrapolated information)
* spending: Average spending per tournament calculated using information from secondary study
* nonlocal: The adjusted spending amount assuming 89% nonlocal tournament participation rate (obtained from secondary study)

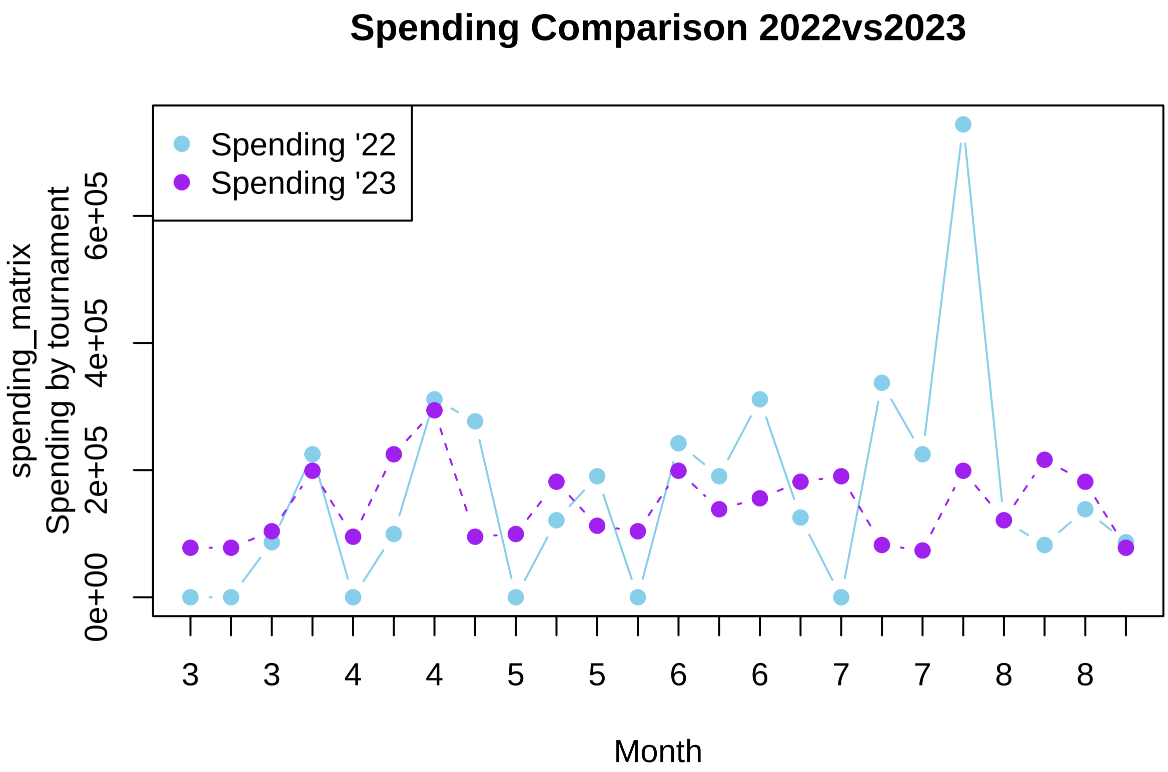
```
##       days         teams            fee              labor         material
##  Min.   :0.0   Min.   : 0.00   Min.   :   0.0   Min.   :   0   Min.   :  0.0
##  1st Qu.:1.0   1st Qu.:16.75   1st Qu.: 600.0   1st Qu.:1326   1st Qu.:219.0
##  Median :1.0   Median :22.00   Median : 600.0   Median :1326   Median :219.0
##  Mean   :1.4   Mean   :19.93   Mean   : 643.2   Mean   :2124   Mean   :272.2
##  3rd Qu.:2.0   3rd Qu.:26.00   3rd Qu.: 870.0   3rd Qu.:3106   3rd Qu.:369.0
##  Max.   :4.0   Max.   :43.00   Max.   :1273.0   Max.   :7000   Max.   :700.0
##     combined        month            players          spending
##  Min.   :   0   Min.   : 3.000   Min.   :  0.0   Min.   :     0
##  1st Qu.:1545   1st Qu.: 4.000   1st Qu.:351.8   1st Qu.: 77868
##  Median :1545   Median : 6.000   Median :462.0   Median :121128
##  Mean   :2396   Mean   : 6.317   Mean   :418.6   Mean   :142314
##  3rd Qu.:3475   3rd Qu.: 8.000   3rd Qu.:546.0   3rd Qu.:192507
##  Max.   :7700   Max.   :11.000   Max.   :903.0   Max.   :744072
##     nonlocal        ...11          ...12
##  Min.   :     0   Mode:logical   Mode:logical
##  1st Qu.: 69303   NA's:60        NA's:60
##  Median :107804
##  Mean   :126659
##  3rd Qu.:171331
##  Max.   :662224
##  Note: Cells 2-36 are 2022 data, 37-64 2023
##  Mode:logical
##  NA's:60
##
##
##
##
```

From a glance at the summary statistics for the information, we can gain some insight into the behaviour at the parks. For instance, we can see that the median number of days for park tournaments is 1 day. We can also see that the mean number of teams is ~19.93 per tournament, and that they were charged on average 643 dollars to participate in the tournament. When looking at player counts we can see that on average each tournament brings around 419 players to the region. Lastly, we can see from data provided, the average initial estimates for expenditures put the average at around $142,314 per tournament. This table as configured includes data for both 2022 and YTD 2023. When 2021 data is provided from the park the data will be added to the table and used in any additional analysis.

Here we can see a graph that compares the two years of data by month:

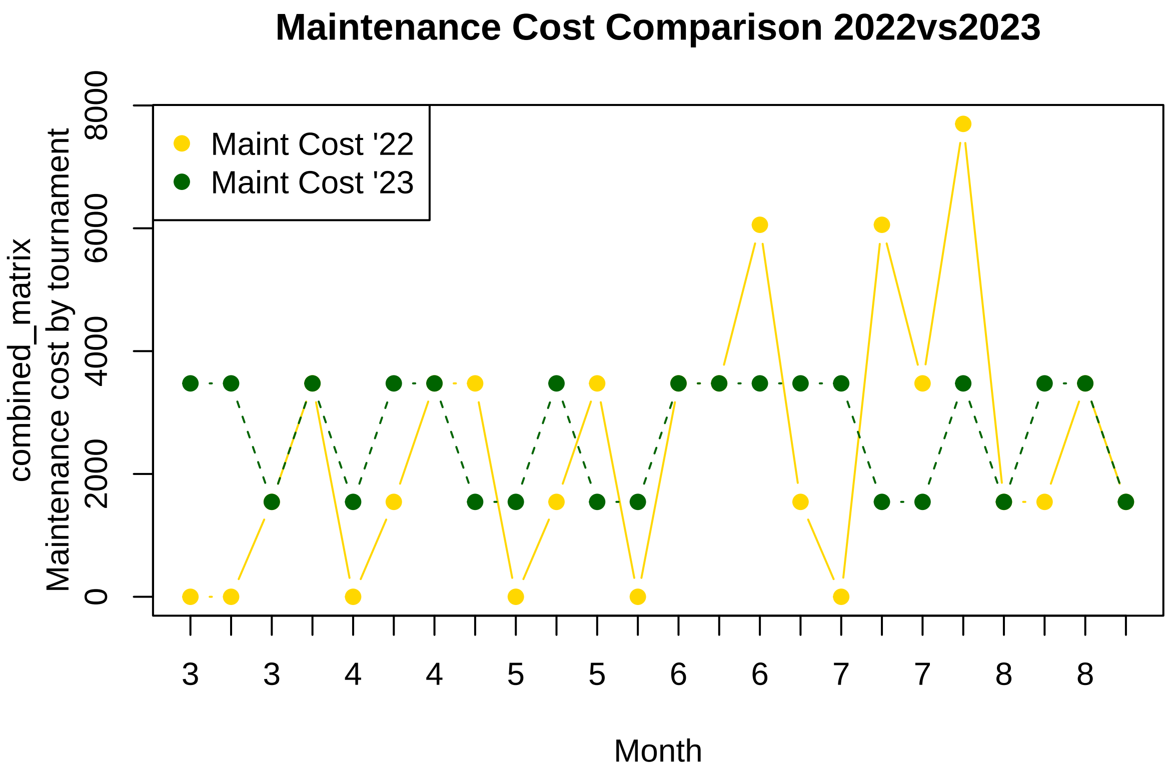


Looking at the graph, we can see that the lines tend to follow similar patterns, remaining fairly constant throughout the season with peaks at the beginning and end. This coincides with the scheduling of baseball/softball, with the start of the season usually being late march and the end of the season being in August/September. Here we can notice that the demand for the fields seems to be fairly constant throughout the baseball/softball season. The zero player counts that are observable are due to rain-outs, in which the tournament was cancelled due to unplayable conditions. The numbers are a little lower for the second half of the season for '23, however any suggestion as to why this has occurred is beyond the scope of this report. Looking at additional visualizations, we can likewise compare spending by year.



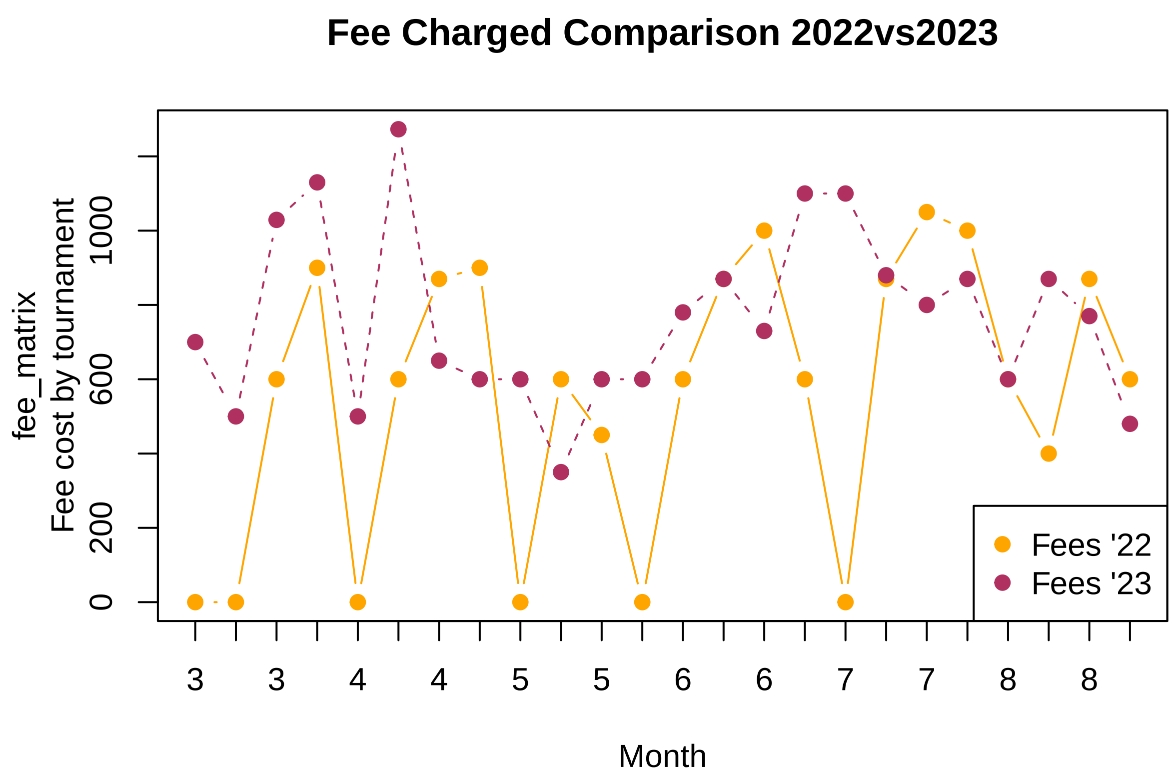
Unsurprisingly we can see that the spending per tournament closely mimics that of the number of player per tournament, intuitively suggesting that the more players there are in a given tournament the more spending occurs in the region. We can also see, again, rather intuitively, that if the tournament does not take place then there is no spending in the region because there are no players in the region. We can also observe that on the average, spending remains fairly constant throughout the season after an initial peak at the beginning of the season (relatively speaking).

We can also look at material cost comparisons between year, though it should be noted that all material cost are essentially constant so this really only shows the difference in tournament type(1,2,3 day) over the two years more than anything else. If more data was available for per tournament cost then a more robust graph could be made with this particular data.



Looking at the maintenance cost comparisons, we see the expected behaviour, where the most we can pull from this is how many of each type of tournament occurred. This is known due to the cost associated with each tournament, where the 1 day tournaments cost ~2000, the 2 day tournaments cost ~3800, the 3 day tournaments cost ~6000, and the 4 day tournament is calculated to have cost ~8000. We can see that 2023 appears to have had a greater number of 2 day tournaments when compared to 2022. There have been no large peaks with a four day tournament so far in 2023.

Lastly, we can look at the fees charged to the tournament directors between the two years.



Looking at the fee comparison between 2022 and 2023, we can see that 2023 actually seems to trend somewhat higher in the average price charged to the tournament directors, relative to the same month. Overall the difference is:

```
## [1] 554
```

```
## [1] 765.9167
```

```
## [1] 211.9167
```

The average for 2022 is 554 dollars and the average for 2023 is 766. This is a 212 dollar increase over the previous year. This is inline with expected cost increases and adjusting pricing to accurately reflect the cost of maintenance.

Part 2 The Secondary Studies

Ok, so no we have some numbers, but now what do we do with them? As we are not well versed in economic impact estimation, we turned to some secondary studies. The most important of these has been "Inexpensively Estimating the Economic Impact of Sports Tourism Programs in Small American Cities" by Brewer and Freeman. This study has been essential to our progress for a multitude of reasons. First and foremost it summarizes the steps required to perform an analysis using its methods, and this summary is clearly laid out. Secondly, it gives us additional information to use in our own study. Third, it has highlighted the need for a certain degree of reliance upon secondary studies to complete the analysis as a full fledged analysis independent of these studies would be both time consuming and expensive. And fourth, because the city of interest in the study is of a similar size and population density as Johnson City, making it almost like a sister city, and as such a great blueprint for creating an analysis for a small city.

The required inputs for this method of analysis are few in number. The first being the number of tourist. This means we need to be able to calculate or estimate the total number of attendees at each tournament, as expenditures often increase as the number of party members increase. However, just as important as the number of attendees itself, we need to know how many of those attendees are non-local. This is because in economic analysis, we cannot count money that is already circulating in the region. This means that if a tournament where to take place at Winged Deer Park, but all the attendees originated from Johnson City, then there would be no measurable impact from the tournament on the local economy because the money would have been spent in the economy either way. For this reason, we must estimate to some degree of accuracy how many non-local attendees come to each tournament in order to obtain the most accurate impact estimates.

The next input we need using this method is average spending per visitor. Fortunately for us, some amount of this information we can obtain locally through visitJC, an governmental agency that collects data on the region, including tourism data. For events that we do not have data on our for which data cannot be provided, we can rely on secondary studies once we adjust the spending to our region.

Lastly, we need the appropriate multiplier and capture rate.This is where the methods in the study begin to deviate from methods outlined in other studies. Here, the multiplier, instead of being produced by using an Input/Output model, they refer to another study (Chang) in which a multiplier for the tourism industry was calculated that relies on the population for the region in which the analysis is being performed and the population density of the region.

We will explore this further in the preliminary calculations section, but for now, its just important to note that this method does not rely upon the Input/Output tables. In addition to the multiplier, we need the capture rate for this method, which is also obtained via a secondary study.

Knowing all of this we can begin to focus on the deficiencies in our own data set.

For starters, being able to find the total spending per party, which we currently don't have. The values for spending in our current data set come from the tables presented in this study, and it is important to note those tables were generated in 2012. Not only has the cost of living increased in this time, but this is also long enough that changes in consumer spending habits should also be taken into consideration. If necessary we can still use this data, we can just adjust the cost of 2012 dollars to current value. For the sake of presenting the most up to date information though, we will continue to explore local sources of information in the hopes of obtaining local and current data.

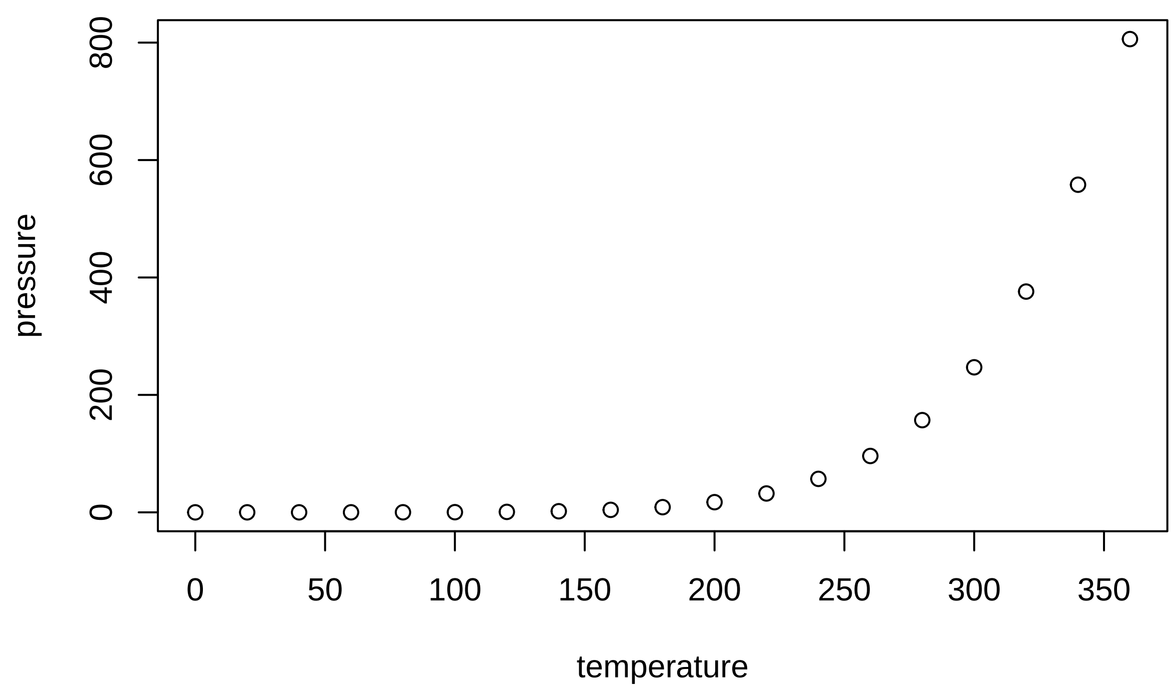
We also need to segment our data by event type. That is easy at the moment as we only have one event type, and that is baseball tournaments. We can bring in local softball information as it becomes available, but for the soccer data, all information will have to be gathered from secondary sources as non exist from local sources.

Once we have this information we then need to find the non-local percentage of attendees. While some of this information may be available from local sources, we will most likely need to find it from a secondary source. This, like most of the other information in the tables needs to be calculated by event type. It is important to segment the information by event type as often the numbers change based on the event. Keeping these separate will allow them to be calculated more accurately and then can be summed together to find the total estimated impact.

Another point of reference has been "The Economic Impact of 30 Sports Tournaments, Festivals, and Spectator Events in Seven US Cities" by Crompton and Lee. In the study, Crompton and Lee bring to light

some important considerations for a study of this type.
One of the main ones being, and rather important to our goals, is that the economic impact on a community when a tournament does not require an overnight stay is small. Typically one should not expect expenditures for these trips to exceed a range of 55 to 100 dollars (in 2000 USD). A follow up to this point is that a large event with a large amount of attendees is unlikely to have a large impact or any impact if 'time-switchers' are not accounted for.
Even more importantly however, the authors outline key principles to adhere to in order to avoid mistakes made in other studies that create overestimates of impact:
* Exclusion of local Residents
* Exclusions of Time Switchers
* Use of Income rather than Sales(Output) Measures
* Careful Interpretation of Employment Measures

You can also embed plots, for example:



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.